

Scooped or Novel? A Temporal Prior-Art Benchmark and a Pilot Ablation of Self-Challenge in Literature-Grounded Research-Idea Assessment

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Abstract

[Pending pilot results.]

1 Introduction

Language-model agents are now asked to judge whether a research idea is new: AI-scientist systems check the novelty of the ideas they generate before executing them [28, 61], and a growing set of tools assesses ideas or submissions against the literature [18, 32, 34, 45, 64]. The failure these tools must avoid is concrete and has been observed: an independent evaluation of the AI Scientist found that “several generated research ideas were incorrectly classified as novel, including well-established concepts such as micro-batching for stochastic gradient descent” [3], and a survey of 24 runnable AI-scientist systems found that only 38% report any novelty-verification method [9].

Two observations motivate this paper. First, evaluations of novelty judgement rest on human labels, which are hard to obtain and contested: LLM judges and domain experts reach opposite conclusions about the novelty of generated research questions [49], LLM judges are swayed by style [21], and even experts find novelty hard to judge [47]. Second, a large diagnostic study of research agents attributes their failures to a missing *metacognitive loop*—“the ability to check what they produced against what they found, revise when it does not hold up”—places the deficit at the model level, and explicitly leaves open “whether orchestration-level interventions can close it” [13]. Novelty assessors in the literature are, by their own descriptions, fixed retrieve-compare-report pipelines; we found none that searches for evidence against its *own* verdict, or that measures what doing so changes (§2).

We address both with a checkable task and a controlled comparison.

A temporal prior-art task. Whether an idea has *already been published* has an objective answer when the realising paper and its date are known. We take recent papers, rewrite each paper’s core idea as a proposal that names nothing identifying, and run an agent on the proposal twice: once with the literature visible up to today (SCOOPED: the right outcome is to find the realising paper and conclude the idea is explored) and once with every work published on or after the realising paper’s date hidden at the literature layer (PREPUB: the paper cannot be found, so an “already explored” verdict must rest on earlier work). We call this Temporal Prior-Art Detection (TPAD).

A controlled ablation of self-challenge. We instrument an open research-investigation harness, ResearchForge, whose controller can (i) raise its own questions that challenge the critique,

each gap and each new direction with a contradiction-seeking literature search, and (ii) revise its critique when such a search weakens a conclusion. Switches turn these off, or replace the adaptive controller with a fixed pipeline, while holding the model, tools, evidence rules and budgets fixed.

Contributions.

1. The TPAD protocol, with anti-leakage controls (a publication-date cutoff enforced at every literature API and on every stored record, first-version full text, disabled code search, a closed-book probe of the agent’s model) and a scorer that reads only the agent’s structured records (§4).
2. An instrumented harness in which self-challenge, critique revision, adaptive control and source-integrity checks are independent switches (§3).
3. A pilot study with DeepSeek-V4.1-Flash on 26 items from papers published in September 2026 (§6). We report it as a pilot: effect estimates with uncertainty, failure analysis, and what a full study needs.

We release the harness, items, run traces and scoring code.

2 Related Work

AI-scientist systems and their audits. End-to-end systems generate ideas, run experiments and write papers [15, 26, 28–30, 41, 42, 53, 61, 67]. Several keep a persistent structured state (Kosmos’s “structured world model” [29]; a “persistent world state” in [57]) and trace statements to sources, as our harness does. Independent audits report weak novelty assessment [3], hypotheses that fail against null models [33], and a verification gap between what systems claim and what reviewers can check [9]. Process-level evaluations find that agents do not revise conclusions against their own evidence [13], reach right answers through wrong mechanisms [11], and differ in feedback control across runs [25]; ARAC scores research-process alignment [8].

Novelty and idea assessment. Benchmarks of novelty judgement use human labels: RINoBench [43], NovBench [58], and NovGauge, which finds that over 70% of correct positive judgements cite evidence that does not support them [63]. LLM judges show a “novelty mirage” [49], a medium-novelty bias [44] and style sensitivity [21]; they fall short of expert preferences [6]. Assessment systems retrieve and compare [16, 18, 32, 34, 45, 46, 62, 64]; ResearchStudio-Idea includes a prior-art collision check for its own generated ideas [66]. Idea generators iterate against literature [2, 22, 24, 35, 54]; human studies show that ideas judged novel lose their advantage after execution [47, 48]. Lit2Test asks proposals to precommit a falsifying outcome [56].

Temporal evaluation. ForeSci hides post-cutoff papers to evaluate forward-looking research judgement and finds “evidence-decision decoupling” [52]; Reconstruction recovers a paper’s idea from its pre-publication bibliography under an anti-leakage protocol [5]; AgentIdeaBench compares static and active exploration for ideation [31]. Search-time contamination can inflate agent benchmark scores [55]. TPAD differs in its target: not recovering or forecasting an idea, but recognising that a given idea is already realised, with the realising paper as checkable evidence.

Falsification and belief revision. POPPER validates hypotheses by sequential falsification *experiments* with error control [19]; a position paper argues for falsification-first agentic science [12]. LLMs show confirmation bias that prompting partly reduces [20], and they fail to retract

conclusions after weakening evidence [40]. Our intervention is literature-level: the agent searches for published work that contradicts its own conclusions.

Evidence and source integrity. Cited synthesis [1, 50] and citation evaluation [14] are established; citation URLs and BibTeX entries are frequently wrong [36, 37, 39], errors originate at orchestrators [17], and claim-level auditability has been proposed as a design target [38]. Upstream, 12% of sampled OpenAlex abstracts have integrity issues [23], and one injected misleading document raises deep-research agents’ false-conclusion adoption from 0% to 54.7% [69]. EviBound gates experiment claims on machine-checkable artefacts [4].

Search control. Deep-research agents stagnate and stop poorly [27, 51, 65]; evidence-sufficiency criteria reduce premature stopping [7]. Benchmarks target reports and QA [10, 59, 68] or literature discovery [60].

3 The Harness

ResearchForge runs an agent (here through DeepSeek Harness, `dsh`, in headless mode) against a set of typed tools exposed over MCP. The model never writes files directly: every finding is a tool call whose arguments are validated and stored as a record in a per-project workspace. A controller outside the model chooses the next action; each action is one agent turn.

Records and evidence rules. Records include retrieved papers (merged across arXiv, OpenAlex, Semantic Scholar and Crossref), paper analyses, claims of kind *evidence*, *inference*, *hypothesis* or *assumption*, a landscape, a critique, gaps, uncertainties, directions and experiment plans. A claim may cite only papers retrieved in the same investigation. *Direct* support requires a quote of at least 20 characters that is found verbatim (after Unicode, whitespace and case normalisation; ellipses allowed between fragments) in the stored abstract or the fetched full text; otherwise the tool call is rejected with an explanation. Hypotheses and assumptions never count as grounded.

Controller. Each iteration the controller (i) raises its own questions, (ii) picks one action, and (iii) measures what changed. A coverage tier runs first: formalize the idea, plan, search until minimum coverage, read, synthesise a landscape, critique. After that, every open uncertainty contributes candidate actions scored by

$$\text{score} = \text{importance} \times \text{expected gain} \times \text{relevance} \times \text{evidence deficiency} / \text{cost},$$

where expected gain blends a prior per action with the information gain the action actually produced earlier in the run. Information gain counts new relevant papers, analyses, verified claims, resolved uncertainties, contradicted conclusions and new structural records. Investigation closes when the last three steps produced no gain or the best candidate falls below a threshold; hard budgets are safety stops. The final *research decision* is derived by rule from the records, not written by the model: `ALREADY_WELL_EXPLORED` if the critique records substantial overlap or the direction is rejected; otherwise `INSUFFICIENT_EVIDENCE` if high-importance questions remain unresolved; `NEEDS_MODIFICATION` if the direction changed or the critique judged the idea weak; otherwise `PROMISING`.

Self-challenge. When a critique, a gap or a new direction is recorded, the controller raises a high-importance question that asks whether published work contradicts it, to be investigated by a contradiction-seeking search (`CHALLENGE` or `INVESTIGATE_GAP`). The agent records a verdict (*holds*, *weakened*, *refuted*, *inconclusive*); *weakened* and *refuted* require contradicting papers that were actually retrieved, and they raise a refinement question.

Critique revision. If a challenge weakens or refutes a conclusion after the current critique was recorded, the controller re-runs the critique before anything else, so the final assessment never predates evidence against it.

Source integrity. Bibliographic APIs sometimes attach the wrong abstract to a record; we observed OpenAlex serving another system’s abstract for H2O (arXiv 2306.14048). When two sources’ abstracts share under 20% of their distinctive words, the one that fits the title is kept and the record is flagged; an abstract that never names the method in a “Name: ...” title is flagged. A flagged abstract cannot supply direct evidence; the quote must be found in full text.

Switches. The components above are independent settings: `challenge_conclusions`, `recritique_on_contradiction`, `integrity_checks`, and `controller_mode`, whose `pipeline` value runs the coverage tier once, then refine, plan and finalise, with no uncertainty-driven investigation and no self-raised questions. All other behaviour—tools, prompts for shared actions, evidence rules, the decision rule and budgets—is identical across systems.

4 Temporal Prior-Art Detection

4.1 Items

An item is built from a *realising paper* P : a recent arXiv paper with a method contribution. It contains

- an *idea*: an 80–150-word proposal stating P ’s problem and approach as a plan, written from P ’s abstract by a model other than the agent’s (`gpt-oss:120b`), instructed not to use P ’s method name, coined acronyms, title phrases, numbers or results;
- P ’s identity and first-version date $d(P)$;
- up to three *prior works*: P ’s references (from Semantic Scholar) dated before P , ranked by distinctive-word overlap with P ’s abstract.

Each draft item records an audit trail: the 5-gram overlap of the idea with P ’s abstract, a check that P ’s method name does not occur in the idea, and a *closed-book probe* in which the agent’s own model is asked, without tools, whether it knows a paper realising the idea. Items whose idea leaks P ’s name, whose overlap exceeds 0.10, or whose probe names P are excluded. Each idea was compared with its realising abstract, for fidelity and for identifying strings, by an LLM assistant (Claude) working for the author; no item has yet been checked by an independent human annotator (§7.1).

4.2 Conditions

Each item is run twice.

scooped. The literature is visible up to the run date. The realising paper exists and is retrievable; the correct outcome is to find it and conclude that the idea is already explored.

prepub. Every work published on or after $d(P)$ is hidden. P cannot be retrieved, so a verdict that the idea is already explored must be justified by earlier work.

Enforcing the cutoff. The cutoff is applied at the literature layer, never told to the model: (i) it is sent to every API (arXiv `submittedDate`, OpenAlex `to_publication_date`, Semantic Scholar `publicationDateOrYear`, Crossref `until-pub-date`); (ii) every record is filtered again before storage, using the earliest date any source reports, and a record known only by year is kept only if the year is earlier than the cutoff’s; (iii) arXiv full text is fetched as version 1, so later revisions cannot leak newer work; (iv) code search is disabled; (v) hidden records are logged. A PREPUB run in which P nevertheless appears in the workspace is marked *leaked* and excluded.

4.3 Scoring

All scores are computed from the workspace records, not from report prose. The outcomes were fixed after one calibration run and before any pilot run (§5).

Primary.

Realising-paper evidence (SCOOPED): P was retrieved, and P is cited in at least one claim with verified evidence. This is detection *for the right reason*.

Verdict discrimination (per item): the research decision is `ALREADY_WELL_EXPLORED` under SCOOPED and not under PREPUB. A verdict that does not change when P is hidden cannot be attributed to P .

Secondary. Verdict-only detection (`ALREADY_WELL_EXPLORED`, or `NEEDS_MODIFICATION` with P cited, under SCOOPED); false-scoop candidates (`ALREADY_WELL_EXPLORED` under PREPUB, which may be justified by earlier work and is inspected case by case); indiscriminate scepticism (`ALREADY_WELL_EXPLORED` under both conditions); prior-work recall (share of the item’s prior works retrieved); cost and process (agent turns, tool calls, tokens, searches, papers retrieved and read, challenges run, contradictions found).

5 Experimental Setup

Items. Candidate realising papers were the 300 most recent arXiv submissions (cs.CL, cs.LG, cs.IR, cs.AI) matching “we propose” in the abstract, submitted 7–10 September 2026, all indexed by Semantic Scholar. From these, an LLM assistant (Claude) working for the author selected 26 method papers in NLP, IR and ML with an idea statable without naming the paper (not benchmarks, surveys, datasets or position papers), for topical diversity. After the exclusion rules of §4, 26 items remained (Appendix A). All realising papers post-date the reported knowledge cutoff of DeepSeek-V4 Flash (April 2026); the closed-book probe additionally checks each item against the agent’s model.

Model and harness. The agent is DeepSeek-V4.1-Flash served by Ollama Cloud through its OpenAI-compatible API, driven by DeepSeek Harness `0.1.7-rc.2` in headless mode with file, shell, web and sub-agent tools disabled, so the only tools are the harness’s literature and record tools. Literature comes from arXiv, OpenAlex, Semantic Scholar and Crossref through a response cache shared by all runs, so that systems compared on an item see identical API responses whenever they issue identical queries. Semantic Scholar was used without an API key and rate-limited many

requests; failed source calls are recorded and the search proceeds with the other sources, for all systems alike.

Systems. FULL: the adaptive controller with self-challenge and critique revision. NOCHAL: identical, with self-challenge off (the controller raises no challenge questions, so critique revision also never triggers). PIPE: the fixed pipeline (coverage tier, one critique, refine, plan, finalise). All share the same hard caps: 20 agent turns, 250 tool calls, 14 searches, 90 papers and 45 minutes per run.

Calibration. One run (FULL, SCOOPED, item 2609.11572) was made before the pilot to set budgets; it is excluded from all analyses. In 26 turns (30 minutes) it retrieved 100 papers but not the realising paper, which is retrievable by its title through arXiv and OpenAlex, and its critique nonetheless recorded substantial overlap with earlier work; its last 15 turns produced repeated experiment plans. The first observation is why verdict-only detection is secondary and verdict discrimination and realising-paper evidence are primary; the second is why the turn cap is 20.

Runs and statistics. Each item is run under both conditions with each system, once (the model is sampled at the provider’s default temperature, so a run is one draw). Comparisons are paired by item. For binary outcomes we report rates with Wilson 95% intervals and exact McNemar tests on discordant pairs; for costs, medians and paired bootstrap 95% intervals of the mean difference (10,000 resamples). With 26 items the study is powered only for large effects; we treat it as a pilot and report effect sizes with their uncertainty rather than significance alone.

6 Results

[Pending.]

7 Discussion

7.1 Limitations

[Pending.]

References

- [1] Akari Asai, Jacqueline He, Rulin Shao, Wei-Jia Shi, Amanpreet Singh, J. Chang, Kyle Lo, Luca Soldaini, Sergey Feldman, Mike D’Arcy, David Wadden, Matt Latzke, Min-Yang Tian, Pan Ji, Shengyan Liu, Hao Tong, Bohao Wu, Yanyu Xiong, Luke S. Zettlemoyer, Graham Neubig, Daniel S. Weld, Doug Downey, Wen tau Yih, Pang Wei Koh, and Hanna Hajishirzi. Openscholar: Synthesizing scientific literature with retrieval-augmented lms. *arXiv preprint arXiv:2411.14199*, 2024. URL <https://arxiv.org/abs/2411.14199>.
- [2] Jinheon Baek, S. Jauhar, Silviu Cucerzan, and Sung Ju Hwang. Researchagent: Iterative research idea generation over scientific literature with large language models. *arXiv preprint arXiv:2404.07738*, 2024. URL <https://arxiv.org/abs/2404.07738>.
- [3] Joeran Beel, Min-Yen Kan, and Moritz Baumgart. Evaluating sakana’s ai scientist: Bold claims, mixed results, and a promising future? *arXiv preprint arXiv:2502.14297*, 2025. URL <https://arxiv.org/abs/2502.14297>.

- [4] Ruiying Chen. Evidence-bound autonomous research (evibound): A governance framework for eliminating false claims. *arXiv preprint arXiv:2511.05524*, 2025. URL <https://arxiv.org/abs/2511.05524>.
- [5] Shao-Long Chen, Yanlin Fei, Nazhou Liu, Xinmiao Yu, Lei Li, Rahul Thapa, M. Ciobanu, N. Singh, Q. Mao, and R. Das. Reconstruction: A blind benchmark for recovering research ideas from pre-publication bibliographies. *arXiv preprint arXiv:2608.16645*, 2026. URL <https://arxiv.org/abs/2608.16645>.
- [6] Zhi-Yu Chen, Keyu Zhao, Ji-Gao Fu, Dong Liang, Yan-Biao Wu, Jiaoyang Li, Haidong Xue, Xin-Hua Zeng, Yuan-Yi Zhen, Fengli Xu, and Yong Li. Ideation arena: Evaluating llm generated research ideas with battle-style human expert assessment. *arXiv preprint arXiv:2608.29696*, 2026. URL <https://arxiv.org/abs/2608.29696>.
- [7] Prafulla Kumar Choubey, Kung-Hsiang Huang, P. Venkit, Jia-Xin Zhang, Vaibhav Vats, Yu Li, Xiangyu Peng, and Chien-Sheng Wu. Dont stop early: Scalable enterprise deep research with controlled information flow and evidence-aware termination. *arXiv preprint arXiv:2604.24978*, 2026. URL <https://arxiv.org/abs/2604.24978>.
- [8] Jiale Cui, Yue-Yao Yuan, Kai-Xi Zhong, Xiao-Gang Xu, Jiafei Wu, and Zhe Liu. Arac: Benchmarking auto-research’s alignment and completeness on end-to-end researchs. *arXiv preprint arXiv:2608.12788*, 2026. URL <https://arxiv.org/abs/2608.12788>.
- [9] Tianyu Ding, Aditya Nannapaneni, Bingfan Liu, and Ling Zhang. Autonomous research agents: A survey of ai scientists and the verification gap. *arXiv preprint arXiv:2608.05179*, 2026. URL <https://arxiv.org/abs/2608.05179>.
- [10] Ming-Xuan Du, Benfeng Xu, Chiwei Zhu, Xiaorui Wang, and Zhendong Mao. Deepresearch bench: A comprehensive benchmark for deep research agents. *arXiv preprint arXiv:2506.11763*, 2025. URL <https://arxiv.org/abs/2506.11763>.
- [11] S. Eulig. Position: Correct answer, wrong mechanism - when ai scientists defend general claims their own data contradicts. *arXiv preprint arXiv:2606.23175*, 2026. URL <https://arxiv.org/abs/2606.23175>.
- [12] Dionizije Fa and Marko uljak. Sound agentic science requires adversarial experiments. *arXiv preprint arXiv:2604.22080*, 2026. URL <https://arxiv.org/abs/2604.22080>.
- [13] Yanlin Fei, Nazhou Liu, Xinmiao Yu, Shao-Long Chen, Lei Li, Rahul Thapa, M. Ciobanu, N. Singh, Q. Mao, and R. Das. How do agents fail on autoresearch: End-to-end diagnostic evaluation on 100 real-world frontier research tasks. *arXiv preprint arXiv:2608.14905*, 2026. URL <https://arxiv.org/abs/2608.14905>.
- [14] Tianyu Gao, Howard Yen, Jia-Tong Yu, and Danqi Chen. Enabling large language models to generate text with citations. *arXiv preprint arXiv:2305.14627*, 2023. URL <https://arxiv.org/abs/2305.14627>.
- [15] JuraJ Gottweis, Wei-Hung Weng, A. Daryin, Tao Tu, Petar Sirkovic, Artiom Myaskovsky, Grzegorz Glowaty, Felix Weissenberger, Alessio Orlandi, D. Popovici, Anil Palepu, Keran Rong, Ryutaro Tanno, Khaled Saab, Fan Zhang, Jacob Blum, A. Carroll, Kavita Kulkarni, Nenad Tomaev, Dina Zverinski, Ivor Rendulic, Elahe Vedadi, Florian Hasler, Luka Rimanic, Marina Boia, Ivan Budiselic, Ben Feinstein, M. Bellaïche, Tom Sheffer, Jan Freyberg, Jeremy Ratcliff,

- Ottavia Bertolli, Katherine Chou, Avinatan Hassidim, Burak Gokturk, Amin Vahdat, Yuan Guan, Vikram Dhillon, E. D. Vaishnav, Byron Lee, Tiago R D Costa, J. Penadés, Gary Peltz, Y. Matias, James Manyika, Demis Hassabis, Yunhan Xu, Pushmeet Kohli, Annalisa Pawlosky, A. Karthikesalingam, and Vivek Natarajan. Accelerating scientific discovery with co-scientist. *arXiv preprint arXiv:2502.18864*, 2025. URL <https://arxiv.org/abs/2502.18864>.
- [16] Shoulin Han. Novelty-aware agentic retrieval: Comparing research contributions through structured multi-step reasoning. *arXiv preprint arXiv:2606.22151*, 2026. URL <https://arxiv.org/abs/2606.22151>.
- [17] Eran Hirsch, David Wan, Han Wang, Elias Stengel-Eskin, Mohit Bansal, and Ido Dagan. Who is the agent to blame? localizing faithfulness and citation mistakes in agentic deep research. *arXiv preprint arXiv:2608.24306*, 2026. URL <https://arxiv.org/abs/2608.24306>.
- [18] Jiajun Hou, Hexuan Deng, Wenxiang Jiao, Xue-Bo Liu, Xiaopeng Ke, Derek F. Wong, and Min Zhang. Memonoveltyagent: A historical research memory-aware agent workflow for paper novelty assessment. *arXiv preprint arXiv:2603.20884*, 2026. URL <https://arxiv.org/abs/2603.20884>.
- [19] Kexin Huang, Ying Jin, Ryan Li, Michael Y. Li, Emmanuel J. Candès, and J. Leskovec. Automated hypothesis validation with agentic sequential falsifications. *arXiv preprint arXiv:2502.09858*, 2025. URL <https://arxiv.org/abs/2502.09858>.
- [20] Ayush Jhaveri, Anthony GX-Chen, Ilia Sucholutsky, and Eunsol Choi. Failing to falsify: Evaluating and mitigating confirmation bias in language models. *arXiv preprint arXiv:2604.02485*, 2026. URL <https://arxiv.org/abs/2604.02485>.
- [21] Feng-Xian Ji, Yuke Li, Jing-Pu Yang, Juanfan Wu, Fan Zhang, Zhexuan Cui, Yuansheng Xie, Min Peng, Qianqian Xie, Xiuying Chen, and Zhuo-Han Xie. Style wins, substance loses: A diagnosis of llm-as-judge in idea generation. *arXiv preprint arXiv:2608.01666*, 2026. URL <https://arxiv.org/abs/2608.01666>.
- [22] Yujing Ke, K. George, Kathan Pandya, D. Blumenthal, M. Sprang, Gerrit GroSSmann, Sebastian J. Vollmer, and D. A. Selby. Biodisco: Multi-agent hypothesis generation with dual-mode evidence, iterative feedback and temporal evaluation. *arXiv preprint arXiv:2508.01285*, 2025. URL <https://arxiv.org/abs/2508.01285>.
- [23] Seorin Kim, Vincent Holst, and Vincent Ginis. One in eight openalex abstracts has integrity issues. *arXiv preprint arXiv:2605.20168*, 2026. URL <https://arxiv.org/abs/2605.20168>.
- [24] Long Li, Wei-Wen Xu, Jiayan Guo, Ruochen Zhao, Xingxuan Li, Yuqian Yuan, Bo-Qiang Zhang, Yuming Jiang, Yifei Xin, Ronghao Dang, Deli Zhao, Yu Rong, Tian Feng, and L. Bing. Chain of ideas: Revolutionizing research via novel idea development with llm agents. *arXiv preprint arXiv:2410.13185*, 2024. URL <https://arxiv.org/abs/2410.13185>.
- [25] Yi-Wei Li, Wanli Yang, He-Xiang Tan, Xiangzhou Huang, Zhengyu Chen, Zi-Ran Li, Borun Chen, Shanglin Lei, Huaisheng Zhu, Haokun Tian, Fei Sun, Xunliang Cai, and Jingang Wang. Beyond final scores: A systematic evaluation of agents for long-horizon ai research and development. *arXiv preprint arXiv:2608.13417*, 2026. URL <https://arxiv.org/abs/2608.13417>.
- [26] Yu Li, Chenyang Shao, Xinyang Liu, Ruotong Zhao, Peijie Liu, Hongyuan Su, Zhi-Bin Chen, Qinglong Yang, Anjie Xu, Yi Fang, Qingbin Zeng, Tianxing Li, Jing-Bo Xu, Fengli Xu, Yong

- Li, and Tie-Yan Liu. Autosota: An end-to-end automated research system for state-of-the-art ai model discovery. *arXiv preprint arXiv:2604.05550*, 2026. URL <https://arxiv.org/abs/2604.05550>.
- [27] Qi Liu, Jia-Xin Mao, Fengbin Zhu, and Tat-Seng Chua. Diagnosing search behavior and failure modes in long-horizon search agents. *arXiv preprint arXiv:2608.01913*, 2026. URL <https://arxiv.org/abs/2608.01913>.
- [28] Chris Lu, Cong Lu, R. Lange, J. Foerster, Jeff Clune, and David Ha. The ai scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024. URL <https://arxiv.org/abs/2408.06292>.
- [29] L. Mitchener, Angela Yiu, Benjamin Chang, M. Bourdenx, Tyler Nadolski, Arvis Sulovari, E. Landsness, Dániel L. Barabási, Siddharth Narayanan, Nicky Evans, S. Reddy, M. Foiani, Aizad Kamal, Leah P. Shriver, F. Cao, A. Wassie, Jon M. Laurent, Edwin Melville-Green, Mayk Caldas, Albert Bou, Kaleigh F. Roberts, Sladjana Zagorac, Timothy C. Orr, Miranda E. Orr, K. Zvezdaryk, A. Ghareeb, L. McCoy, B. Gomes, Euan A. Ashley, K. Duff, T. Buonassisi, Tom Rainforth, R. Bateman, Michael Skarlinski, Samuel G. Rodrigues, Michaela M. Hinks, and Andrew D. White. Kosmos: An ai scientist for autonomous discovery. *arXiv preprint arXiv:2511.02824*, 2025. URL <https://arxiv.org/abs/2511.02824>.
- [30] Atsuyuki Miyai, M. Toyooka, Takashi Otonari, Zaiying Zhao, and Kiyoharu Aizawa. Jr. ai scientist and its risk report: Autonomous scientific exploration from a baseline paper. *arXiv preprint arXiv:2511.04583*, 2025. URL <https://arxiv.org/abs/2511.04583>.
- [31] Yunxiang Mo, Tianshi ZHENG, Yi-Sen Gao, Rui Wang, Newt Nguyen Kim Hue Nam, K. Tam, Jiabin Bai, Yangqiu Song, Ginny Y. Wong, and Simon See. Agentideabench: Benchmarking scientific ideation in the agent era. *arXiv preprint arXiv:2609.07611*, 2026. URL <https://arxiv.org/abs/2609.07611>.
- [32] Hanane Nour Moussa, Patrick Queiroz Da Silva, Daniel Adu-Ampratwum, Alyson East, Zi-Tong Lu, Nikki A. Puccetti, Mingyi Xue, Huan Sun, Bodhisattwa Prasad Majumder, and Sachin Kumar. Scholareval: Research idea evaluation grounded in literature. *arXiv preprint arXiv:2510.16234*, 2025. URL <https://arxiv.org/abs/2510.16234>.
- [33] H. Nusrat and Omar Nusrat. When ai does science: Evaluating the autonomous ai scientist kosmos in radiation biology. *arXiv preprint arXiv:2511.13825*, 2025. URL <https://arxiv.org/abs/2511.13825>.
- [34] Shuo-Fei Qiao, Yun-Xiang Wei, Xue-Hai Wang, Bin Wu, Bo-Yang Xue, Ningyu Zhang, Hossein A. Rahmani, Yanshan Wang, Qiang Zhang, Keyan Ding, Jeff Z. Pan, Hua-Jun Chen, and Emine Yilmaz. Innoeval: On research idea evaluation as a knowledge-grounded, multi-perspective reasoning problem. *arXiv preprint arXiv:2602.14367*, 2026. URL <https://arxiv.org/abs/2602.14367>.
- [35] Marissa Radensky, Simra Shahid, Raymond Fok, Pao Siangliulue, Tom Hope, and Daniel S. Weld. Human-llm compound system for scientific ideation through facet recombination and novelty evaluation. *arXiv preprint arXiv:2409.14634*, 2024. URL <https://arxiv.org/abs/2409.14634>.

- [36] D. Rao and Christopher Callison-Burch. Bibtex citation errors in scientific publishing agents: Evaluation and mitigation. *arXiv preprint arXiv:2604.03159*, 2026. URL <https://arxiv.org/abs/2604.03159>.
- [37] D. Rao, Eric Wong, and Christopher Callison-Burch. Detecting and correcting reference hallucinations in commercial llms and deep research agents. *arXiv preprint arXiv:2604.03173*, 2026. URL <https://arxiv.org/abs/2604.03173>.
- [38] Razeen A Rasheed, Somnath Banerjee, Animesh Mukherjee, and Rima Hazra. From fluent to verifiable: Claim-level auditability for deep research agents. *arXiv preprint arXiv:2602.13855*, 2026. URL <https://arxiv.org/abs/2602.13855>.
- [39] Patrik Reizinger and Wieland Brendel. Hallmark: Diagnosing three failure modes in llm citation verifiers. *arXiv preprint arXiv:2607.18360*, 2026. URL <https://arxiv.org/abs/2607.18360>.
- [40] Jayanta Sadhu, S. Shahad, and Kenneth Marino. Derelab: Probing defeasible reasoning and confirmation bias in llms with a generative benchmark. *arXiv preprint arXiv:2608.30413*, 2026. URL <https://arxiv.org/abs/2608.30413>.
- [41] S. Schmidgall and Michael Moor. Agentrxiv: Towards collaborative autonomous research. *arXiv preprint arXiv:2503.18102*, 2025. URL <https://arxiv.org/abs/2503.18102>.
- [42] Samuel Schmidgall, Yu-Sheng Su, Ze Wang, Ximeng Sun, Jia-Lian Wu, Xiaodong Yu, Jiang Liu, Michael Moor, Zi-Cheng Liu, and E. Barsoum. Agent laboratory: Using llm agents as research assistants. *arXiv preprint arXiv:2501.04227*, 2025. URL <https://arxiv.org/abs/2501.04227>.
- [43] Tim Schopf and Michael Farber. Is this idea novel? an automated benchmark for judgment of research ideas. *arXiv preprint arXiv:2603.10303*, 2026. URL <https://arxiv.org/abs/2603.10303>.
- [44] Tim Schopf, Tobias Schreieder, and Akiko Aizawa. Think-probe-respond: Improving large language models as judges of research idea novelty. *arXiv preprint arXiv:2608.25660*, 2026. URL <https://arxiv.org/abs/2608.25660>.
- [45] Simra Shahid, Marissa Radensky, Raymond Fok, Pao Siangliulue, Daniel S. Weld, and Tom Hope. Literature-grounded novelty assessment of scientific ideas. *arXiv preprint arXiv:2506.22026*, 2025. URL <https://arxiv.org/abs/2506.22026>.
- [46] Yue Shen, Minqian Liu, Da-Wei Zhou, and Li-Fu Huang. Navigating ideation space: Decomposed conceptual representations for positioning scientific ideas. *arXiv preprint arXiv:2601.08901*, 2026. URL <https://arxiv.org/abs/2601.08901>.
- [47] Chenglei Si, Diyi Yang, and Tatsunori Hashimoto. Can llms generate novel research ideas? a large-scale human study with 100+ nlp researchers. *arXiv preprint arXiv:2409.04109*, 2024. URL <https://arxiv.org/abs/2409.04109>.
- [48] Chenglei Si, Tatsunori Hashimoto, and Diyi Yang. The ideation-execution gap: Execution outcomes of llm-generated versus human research ideas. *arXiv preprint arXiv:2506.20803*, 2025. URL <https://arxiv.org/abs/2506.20803>.

- [49] Soumitra Sinhaajari, Navonil Majumder, and Soujanya Poria. On the limits of llm-as-judge for scientific novelty assessment. *arXiv preprint arXiv:2606.12071*, 2026. URL <https://arxiv.org/abs/2606.12071>.
- [50] Michael Skarlinski, Sam Cox, Jon M. Laurent, James D. Braza, Michaela M. Hinks, M. Hammerling, Manvitha Ponnampati, Samuel G. Rodrigues, and Andrew D. White. Language agents achieve superhuman synthesis of scientific knowledge. *arXiv preprint arXiv:2409.13740*, 2024. URL <https://arxiv.org/abs/2409.13740>.
- [51] Heydar Soudani, Elisabeth Lingg, Faegheh Hasibi, and Navid Rekabsaz. When deep research agents stagnate: Enhancing reasoning with retrieval-aware agent control. *arXiv preprint arXiv:2608.15191*, 2026. URL <https://arxiv.org/abs/2608.15191>.
- [52] Qiuyu Tian, Hao Yin, Yingce Xia, You-Yong Kong, and Zequn Liu. Foresci: Evaluating llm agents for forward-looking ai research judgment. *arXiv preprint arXiv:2606.00644*, 2026. URL <https://arxiv.org/abs/2606.00644>.
- [53] Cheng-Cheng Wang, Qinhua Xie, Wei He, Jian-Yuan Guo, Shiqi Wang, and Chang Xu. Sibyl-autoresearch: Autonomous research needs self-evolving trial-and-error harnesses, not paper generators. *arXiv preprint arXiv:2605.22343*, 2026. URL <https://arxiv.org/abs/2605.22343>.
- [54] Qingyun Wang, Doug Downey, Heng Ji, and Tom Hope. Scimon: Scientific inspiration machines optimized for novelty. *arXiv preprint arXiv:2305.14259*, 2023. URL <https://arxiv.org/abs/2305.14259>.
- [55] Yong-Jie Wang, Xinyu Crystina Zhang, Kunhong Yao, Zhiwei Zeng, Kai-Song Song, Jun Lin, and Zhi-Qiang Shen. Search-time contamination in deep research agents: Measuring performance inflation in public benchmark evaluation. *arXiv preprint arXiv:2606.05241*, 2026. URL <https://arxiv.org/abs/2606.05241>.
- [56] Zi-Yue Wang, Aomufei Yuan, Yi-Ran Yao, Linli Yao, Hongyao Zuo, Ziwen Gong, Yuanxin Liu, Shi-Cheng Li, Yishuo Cai, Tong Yang, Xu Sun, Xiao-Hui Li, Haoli Bai State Key Laboratory of Multimedia Information Processing, School of Materials Science, Peking University, Tianjin University, Hainan University, and Huawei Technologies. What proves you wrong: Benchmarking language models on falsifiable research ideation. *arXiv preprint arXiv:2608.22948*, 2026. URL <https://arxiv.org/abs/2608.22948>.
- [57] L. Weidener, Marko Brkić, Mihailo R. Jovanovi, Ritvik Singh, Chiara Baccin, E. Ulgac, A. Dobrin, and A. Meduri. Rethinking the ai scientist: Interactive multi-agent workflows for scientific discovery. *arXiv preprint arXiv:2601.12542*, 2026. URL <https://arxiv.org/abs/2601.12542>.
- [58] Wen-Qing Wu, Yi Zhao, Yuzhuo Wang, Siyou Li, Juexi Shao, Yun-Fei Long, and Cheng-Zhi Zhang. Novbench: Evaluating large language models on academic paper novelty assessment. *arXiv preprint arXiv:2604.11543*, 2026. URL <https://arxiv.org/abs/2604.11543>.
- [59] Qianqian Xie, Qing Xiong, He Zhu, Tian Xia, Xue Han, Fanyu Meng, Jiakai Wang, Zhi-Qi Bai, Chengkang Jiang, Zhao-Hui Wang, Yu-Bin Guo, Yu-Qing Wen, JiaYan Mao, Zi Xuan Zhang, Shi-Hao Li, Yang-Hai Wang, Yuxiang Ren, Jun-Lan Feng, and Jia-Heng Liu. Dr3-eval: Towards realistic and reproducible deep research evaluation. *arXiv preprint arXiv:2604.14683*, 2026. URL <https://arxiv.org/abs/2604.14683>.

- [60] Lei Xiong, Kun Luo, Ziyi Xia, Wen-Bo Zhang, Jin-Ge Yao, Zheng Liu, Jing Shao, Jianlyu Chen, Hong-Jin Qian, Xi Yang, Qian Yu, Hao Li, C. Yue, Xiaan Du, Yuyang Wang, Ye-Sheng Liu, Hai-Yu Xu, and Zhi-Cheng Dou. Autoresearchbench: Benchmarking ai agents on complex scientific literature discovery. *arXiv preprint arXiv:2604.25256*, 2026. URL <https://arxiv.org/abs/2604.25256>.
- [61] Yutaro Yamada, R. Lange, Cong Lu, Shengran Hu, Chris Lu, J. Foerster, Jeff Clune, and David Ha. The ai scientist-v2: Workshop-level automated scientific discovery via agentic tree search. *arXiv preprint arXiv:2504.08066*, 2025. URL <https://arxiv.org/abs/2504.08066>.
- [62] Zheng-Xu Yan, Han Li, and Yuming Feng. Noveltyrank: A retrieval-augmented framework for conceptual novelty estimation in ai research. *arXiv preprint arXiv:2512.14738*, 2025. URL <https://arxiv.org/abs/2512.14738>.
- [63] Guo-Qiang Zhang, Ke-Xin Tan, Ming Zhang, Li Ju, Wenqing Jing, Zhong-Han Yue, Jia-Yi Chen, Shi-Qiang Wu, Shao-Hua Liu, Yue Zhang, Yuankai Ying, Yang Shi, Tao Gui, Qi Zhang, and Xuan-Jing Huang. Novgauge: A fine-grained benchmark for diagnosing llms’ capability in paper novelty assessment. *arXiv preprint arXiv:2609.11234*, 2026. URL <https://arxiv.org/abs/2609.11234>.
- [64] Ming Zhang, Ke-Xin Tan, Yue-Yuan Huang, Yu-Jiong Shen, Chunchun Ma, Li Ju, Xinran Zhang, Yu-Hui Wang, Wenqing Jing, Jingyi Deng, Huayu Sha, Binze Hu, Jing-Qi Tong, Changhao Jiang, Yage Geng, Yuankai Ying, Yue Zhang, Zhangyue Yin, Zhi-Heng Xi, Shi-Han Dou, Tao Gui, Qi Zhang, and Xuan-Jing Huang. Opennovelty: An llm-powered agentic system for verifiable scholarly novelty assessment. *arXiv preprint arXiv:2601.01576*, 2026. URL <https://arxiv.org/abs/2601.01576>.
- [65] Wenlin Zhang, Kui-Cai Dong, J. Li, Yingyi Zhang, Xiaopeng Li, Peng-Yue Jia, Yi Wen, De-Rong Xu, Mao-Lin Wang, Yi-Chao Wang, Yong Liu, and Xiangyu Zhao. To search or not to search: Aligning the decision boundary of deep search agents via causal intervention. *arXiv preprint arXiv:2602.03304*, 2026. URL <https://arxiv.org/abs/2602.03304>.
- [66] Qihao Zhao, Yangyu Huang, Yalun Dai, Lingao Xiao, Jianjun Gao, Xin Zhang, Wenshan Wu, Scarlett Li, Yang He, Yan Lu, and Yap Kim Hui. Researchstudio-idea: An evidence-grounded research-ideation skill suite from ml conference outcomes. *arXiv preprint arXiv:2607.04439*, 2026. URL <https://arxiv.org/abs/2607.04439>.
- [67] Y. Zheng, Yu-Xin Wang, Jiahao Lu, Shicheng Fang, Weiyi Wang, Yongzhuo Yang, Bo-Wen Li, Hao Ma, Chengcheng Hu, Bowen Chen, Yang Wang, Huanhui Chen, Yitong Chen, Jiajun Chen, Zhiyuan Li, Yanling Li, Zhuo Yang, Qi-Feng Wu, Jiaying He, Zhijie Jinluo, Xiao-Hua Xu, Yi Feng, Juncheng Qian, Yizhou Chen, Yang Cheng, Tong Zhu, Tianlei Ying, Hongyu Yu, Hong-Jun Xiang, and Xipeng Qiu. Rethinking scientific discovery in the agentic era. *arXiv preprint arXiv:2607.03863*, 2026. URL <https://arxiv.org/abs/2607.03863>.
- [68] J. Zhong, Hao Zhang, Clare Southern, Jeremy Yang, Thomas Wang, K. Jung, Shu Zhang, Denis Yarats, Johnny Ho, and Jerry Ma. Draco: a cross-domain benchmark for deep research accuracy, completeness, and objectivity. *arXiv preprint arXiv:2602.11685*, 2026. URL <https://arxiv.org/abs/2602.11685>.
- [69] Peng-Yu Zhu, Lijun Li, Long-Ping Yang, Sen Su, and Jing Shao. Is deep research reliable? misleading knowledge induces false conclusions. *arXiv preprint arXiv:2607.20891*, 2026. URL <https://arxiv.org/abs/2607.20891>.

A Items

All 26 items. $d(P)$ is the realising paper’s first arXiv version date (the PREPUB cutoff); “Prior” is the number of prior works found among its Semantic Scholar references (0 when the reference list was unavailable, in which case prior-work recall is undefined for the item). Idea texts, prior works and the audit trail of each item are released with the data.

Item	Realising paper	$d(P)$	Prior
2609.12230	Repair Before Reinforce: Context-Augmented Knowledge Graph Reasoning for Multi-Hop Question Answering	2026-09-10	0
2609.11572	TimelyRAG: Semantic-Temporal Hybrid Retrieval for Time-Critical Question Answering in Overlapping-Evolving Documents	2026-09-10	3
2609.11446	Calibration-Aware Uncertainty Cascades for Efficient Heterogeneous Model Collaboration	2026-09-10	3
2609.11414	SWRouter: Similarity-Contractive Window Routing for Multi-Turn Large Language Model Conversations	2026-09-10	3
2609.11393	Beyond Confidence: Stability-Aware Test-Time Adaptation for LLM Reasoning	2026-09-10	3
2609.11192	FlexComp: One Model for Every Ratio in Context Compression	2026-09-10	3
2609.11061	Fork Where the Model Changes Its Mind: Belief-Shift Branching for Tree-Structured Reinforcement Learning	2026-09-10	3
2609.10441	ConvMem: Convolutional Memory for Long-Context Reasoning	2026-09-09	3
2609.10439	Forgetting Only What Matters: Layer-Selective Unlearning toward Robust LLMs	2026-09-09	3
2609.10346	Beyond One-Size-Fits-All: Sample-Adaptive Strategy Routing for Vision Token Pruning in MLLMs	2026-09-09	3
2609.09883	Forward-Free LLM Depth Pruning via Weight Redundancy	2026-09-09	3
2609.09768	Fine-Tuning a KV Cache Concatenation-Aware Model or Re-computing KV Caches? Why Not Both?	2026-09-09	3
2609.09707	Which Tokens Should SFT Actually Learn? A Token-Trimming Perspective on Mathematical Reasoning	2026-09-09	3
2609.09835	HyperTrace: Hypothesis-Based Preference Tracing for Online LLM Personalization	2026-09-09	3
2609.09156	ReCite: Agentic Reasoning for Faithful Citation	2026-09-08	3
2609.08650	Difficulty-Adaptive Tree-Structured Policy Optimization for Expanding Reasoning Coverage in RLVR	2026-09-08	3
2609.08267	Evidence-Aligned Entity Verification for Hallucination Detection in Retrieval-Augmented Generation	2026-09-08	3
2609.08189	Do Dynamic Routers Need Memory? HeRo: History-Aware Routing for Efficient LLM Inference	2026-09-08	3
2609.08273	MemForest: Efficient Agent Memory Management via Event-Tree Partitioning and Progressive Merging	2026-09-08	0
2609.07286	TAD: Token-Adaptive Contrastive Decoding with Confidence-Guided Gating for Hallucination Mitigation in Large Audio-Language Models	2026-09-07	3
2609.07050	Beyond One-Shot Expansion: Contrastive Evidence Exploration for Multi-Hop Retrieval	2026-09-07	3
2609.10060	Reference-Based Bias Detection in LLMs via Relative Representations of Hidden States	2026-09-09	3
2609.09953	Improving Cross-Lingual Token Representations by Adding a Pinch of SALT	2026-09-09	0

Item	Realising paper	$d(P)$	Prior
2609.11655	Musec: MomentUm SpEctral Clipping for Stable Muon-type Training	2026-09-10	3
2609.30289	Not All Memories Are Equal: Hierarchical Collaborative Memory for Validity-Aware Retrieval in LLM Agents	2026-09-09	3
2609.08173	Key Path Identification for Resolving Knowledge Conflicts via SAE-based Steering	2026-09-08	3

B Prompts used to build items

Paraphrase (model `gpt-oss:120b`, temperature 0.2):

Rewrite the core idea of the paper below as a short research proposal (80-150 words) written BEFORE the work was done. Rules: - State the problem and the proposed approach so a researcher could recognise the idea. - Do NOT use the paper’s method name, any acronym it coins, its title words in sequence, author names, numbers, or results. - Do not claim it works; phrase it as a plan ("We propose to ...", "The idea is to ..."). - Use your own wording; do not copy phrases from the abstract. Output only the proposal text. Title: {title} Abstract: {abstract}

Closed-book probe (model `deepseek-v4.1-flash`, the agent’s model, no tools):

Here is a research idea. Do you know of a specific published paper (arXiv or venue) that proposes essentially this idea? If yes, give its exact title and year. If you are not sure, answer NONE. Answer in one line. Idea: {idea}

A probe counts as naming the realising paper if its answer contains the first 40 normalised characters of the paper’s title or the paper’s method name.

C Reproducibility

The harness, the benchmark runner (**researchforge bench**), the item builder and the analysis script are released. Each run directory contains the settings file both the orchestrator and the tool server read, every controller decision with its candidates and observed information gain, the agent’s structured records, the dsh event streams, and the ledger of records hidden by the cutoff. All literature API responses are cached, so scores can be recomputed offline and runs replayed against identical API responses.